Mississippi Valley State University

# CS 341 — Discrete Structures

Practice Set 27

*Proof by Contradiction*

Covers Chartrand & Zhang, Section 3.7

**Status of this set**

**Ungraded practice.** Solutions released with it. Write every proof in complete sentences, and begin each with "Assume, to the contrary, that…"

# Part I: How to Practice This Material

**The template**

1. **Assume the negation** of what you want to prove. For an implication, assume the hypothesis holds *and* the conclusion fails.
2. **Name** anything whose existence you have just assumed.
3. **Derive** a statement that contradicts a known fact or one of your assumptions.
4. **Conclude** that the assumption was false, so the original statement is true.

**When to prefer it**

Reach for contradiction when there is **no implication to reverse** — "$\sqrt{2}$ is irrational," "there is no largest…" If the statement is an implication and the contrapositive works with the same algebra, the contrapositive is cleaner.

**Three mistakes to expect**

1. **Negating incorrectly.** The negation of $P \Rightarrow Q$ is $P \wedge (\sim Q)$, not $(\sim P) \Rightarrow (\sim Q)$.
2. **Never reaching a contradiction.** Say explicitly what is contradicted.
3. **Using contradiction where contrapositive is simpler.** Legal, but longer.

# Part II: Practice Problems

## Section A — "There is no…"

**1.** Prove that there is no largest even integer.

**2.** Prove that there is no largest rational number less than 1.

## Section B — Irrationality

**3.** You may use the fact: *for an integer $n$, if $n^2$ is divisible by 5, then $n$ is divisible by 5.* Prove that $\sqrt{5}$ is irrational.

**4.** Prove: if $a$ is rational and $b$ is irrational, then $a + b$ is irrational.

## Section C — Parity

**5.** Prove by contradiction: let $n \in \mathbb{Z}$; if $3n - 11$ is odd, then $n$ is even.

## Section D — Writing

**6.** In five to eight complete sentences, explain why deriving a contradiction proves the original statement. Name the rule of inference from Week 3 that makes this legitimate.

## Section E — Challenge (optional)

**7.** Prove that $\sqrt{2} + \sqrt{3}$ is irrational.